

H06 Compression → Expansion

Breakouts from volatility compression on NQ futures: research report for the trading decision (cohort members #6 C2839e22b29c0cb17 and #18 C39c52e02a3c54412)

Prepared for the trading decision committee · 23 September 2026 · data through 10 August 2026 · all figures per 1 NQ contract unless stated

Recommendation: DO NOT TRADE either H06 variant. Both failed the preregistered, untouched 2023–2025 holdout, which is the test that counts. H06-B (long only) is negative on every post-freeze measure and should be dropped. H06-A stays in the frozen cohort's passive forward record only. It is not eligible for capital unless the true-forward record from 24 Sep 2026 independently meets the criteria on the last page.

Executive summary

- **What it is.** Buy (sell) NQ when the close breaks the previous 60-minute high (low) after 60 minutes of unusually quiet trading (realised vol $\leq 0.6\times$ its normal level for that time of day). Variant A also requires a strong 5-minute push and trades both directions with a $1\times$ ATR stop. Variant B is long only with a $1.5\times$ ATR stop. There is no profit target: trades exit at the stop or at 16:00.
- **Why it was selected.** The 2019–2022 confirmation test was strong. A made +\$61.0k on 531 trades (PF 1.43, t 2.58), B made +\$60.3k on 435 trades (PF 1.48, t 2.79), both beating 98% of no-edge null worlds (p 0.020). A was rated STRONG and B PROMISING.
- **Why it fails the decision.** In the untouched 2023–25 holdout A lost −\$4.4k (PF 0.97) and B lost −\$9.0k (PF 0.93), both significantly below what confirmation predicted (z −1.78 and −2.46). Both are preregistered **FAILED**. Over 2024–2025 alone: A −\$9.7k on 241 trades (PF 0.91), B −\$11.7k on 214 trades (PF 0.89).
- **The edge in discovery was thin.** Before selection, 2010–2018 gave A only +0.019R per trade (t 0.25 in R) and B +0.049R. The strong numbers came almost entirely from 2019–2022, which the campaign itself flags as an unusually trending, high-volatility period.
- **The compression filter is what stopped working.** In 2019–22 the ‘quiet before the break’ condition improved most neighbouring variants (8 of 9 frozen pairs). In 2023–25 every long or two-sided variant using the frozen 0.6 threshold had PF < 1 (0.79–0.98), while the same breakouts without it were about break-even. The information the strategy relies on did not carry forward.
- **The mitigating evidence is weak and post hoc.** A was positive in R in every holdout year (+32R) and strong in 2026 (+\$26.4k, PF 2.11, 53 trades), and its short side made +84R after 2022. But the 2026 sample is small, the short-side split was found after the results were known, and the preregistered verdict cannot be revised by these observations.

Headline results by research stage

Strategy	Period	Trades	Net P&L	PF	Win %	Net R	Avg R	Max DD	Verdict
H06-A	Discovery 2010–18	742	\$9,162	1.17	32.6%	+14.5	+0.019	\$3,457	PRIMARY
H06-A	Confirmation 2019–22	531	\$61,006	1.43	33.5%	+196.3	+0.370	\$9,376	STRONG
H06-A	Holdout 2023–25	331	−\$4,424	0.97	29.0%	+32.2	+0.097	\$35,340	FAILED
H06-A	2026 audit (Jan–Aug)	53	\$26,378	2.11	43.4%	+39.9	+0.752	\$8,832	SUPPORTED
H06-A	2024–2025 only	241	−\$9,669	0.91	27.8%	+21.0	+0.087	\$35,340	–
H06-A	All post-freeze 2023–26	384	\$21,954	1.14	31.0%	+72.1	+0.188	\$35,340	–
H06-B	Discovery 2010–18	618	\$7,733	1.15	39.6%	+30.0	+0.049	\$5,356	PRIMARY
H06-B	Confirmation 2019–22	435	\$60,290	1.48	46.0%	+122.5	+0.282	\$10,414	PROMISING
H06-B	Holdout 2023–25	303	−\$9,047	0.93	35.6%	−21.7	−0.072	\$22,839	FAILED
H06-B	2026 audit (Jan–Aug)	47	\$3,947	1.13	38.3%	+3.2	+0.068	\$10,033	WEAKER
H06-B	2024–2025 only	214	−\$11,736	0.89	35.5%	−20.3	−0.095	\$20,189	–
H06-B	All post-freeze 2023–26	350	−\$5,100	0.97	36.0%	−18.5	−0.053	\$29,062	–

Stages before the freeze (discovery, confirmation) were used to select the strategy; 2023 onward was never seen before the cohort was frozen on 22 Sep 2026 23:21 UTC. Verdicts are the campaign's preregistered classifications. $R = \text{P\&L}$ divided by the trade's own initial risk.

1. How this strategy was found and tested

The strategy came out of the QuantLab V3 research campaign: 53 preregistered hypothesis families (33,077 specifications) were searched on 2010–2018 NQ/ES 1-minute data. Survivors were frozen and re-tested unchanged on 2019–2022 against 100 synthetic ‘no-edge’ market worlds, clustered into distinct behaviours, and 25 were selected into a frozen cohort. The cohort was then run once on the untouched 2023–2025 holdout, and once on 2026 data to 10 Aug 2026. Nothing was changed after any later stage was seen; that is what makes the post-2022 numbers honest out-of-sample evidence.

Stage	Dates	Role	What could change
Discovery	2010-06-07 → 2018-12-31	search (in-sample)	everything: rules, grid, management
Confirmation	2019 → 2022	out-of-sample test vs null worlds	only which frozen candidates are kept
Final cohort freeze	22 Sep 2026 23:21 UTC	spec hash f5666dac...	nothing after this point
Holdout	2023 → 2025	untouched out-of-sample	nothing (report only)
2026 audit	2026-01-01 → 2026-08-10	secondary, low power	nothing (report only)
True forward	from 2026-09-24	live-market paper record	nothing (forward/run_forward.py)

Reproducibility. For this report every trade from 2010 to 2026 was regenerated with the frozen specification, the stage-gated data loader and the campaign engine. Trade counts and net P&L match the sealed campaign records exactly in all four periods for every strategy covered here.

Why R-multiples matter here. NQ traded near 2,000 in 2010 and near 27,000 in 2026, and the stop is volatility-based, so one trade's dollar risk ranged from roughly \$50 to over \$2,000 on one contract. Dollar totals are therefore dominated by recent, high-volatility years, and a single contract is a very different bet in 2012 than in 2026. We report both dollars (the campaign's preregistered metric) and R ($\text{P\&L} \div \text{initial risk}$), which approximates a book that risks a constant amount per trade. **The official verdicts use dollars; the R view is a descriptive, post-hoc lens and cannot overturn them.**

Hypothesis family

Preregistered hypothesis (H06). A close beyond the previous N-bar high/low starts a directional expansion. Confluence: the N bars before the break were compressed (realised vol over N bars $\div$ its time-of-day norm $\leq$ cth). Optional strength filter: the 5-minute move $\geq 1 \sigma$. Grid: $N \in \{30, 60\}$, $\text{cth} \in \{\text{none}, 0.6, 0.75\}$, $\text{strong} \in \{\text{none}, 1.0\}$, continuation only, both / long / short. The family had 756 specifications in discovery, 38 of them PRIMARY (best t 2.13, median PF 0.92).

Rationale. Volatility clusters and mean-reverts. After an unusually quiet hour, the next range break often coincides with new information or order flow and tends to extend. The ablation tests asked whether adding the compression condition beats the plain breakout. In discovery it raised the average trade in 80% of 312 paired comparisons (median +\$4.26 per trade), and in the 9 pairs where both specs were frozen it improved 8 in confirmation. The strength filter added less (85% of pairs in discovery, 50% in confirmation).

Behaviour clustering. The two variants belong to different behaviour clusters: #3, 13 members and 1 entry rule, STRONG; and #2, 19 members and 2 entry rules, PROMISING. Their daily P&L correlated 0.43 in the holdout, so they are largely the same bet. B had one of only three family-wise-significant confirmation t-stats in the whole cohort (p 0.099). That did not protect it in the holdout.

2. H06-A (#6, long + short): specification

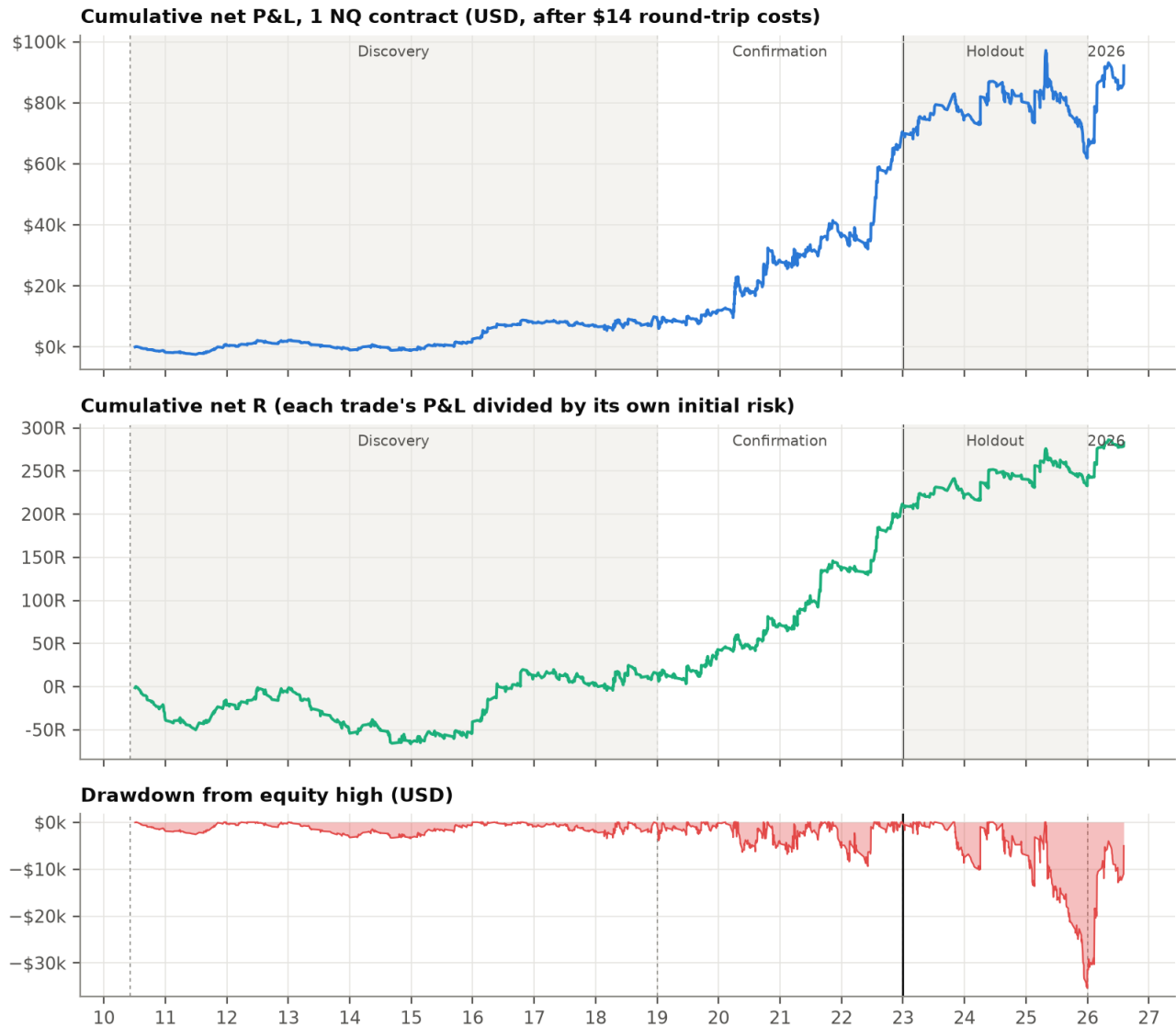
Field	Frozen value
Strategy ID	C2839e22b29c0cb17
Entry rule ID	Efc242ab81ddb80b3
Family / version	v3_h06_compression_expansion (v3.0)
Instrument	NQ (E-mini Nasdaq-100 futures, \$20 per point), front contract, 1-minute bars
Direction	both
Mode	continuation
Long trigger	the close breaks above the previous 60-bar high; the previous 60 bars' realised vol was $\leq 0.6 \times$ its time-of-day norm; the 5-minute move ≥ 1.0 sigma units
Short trigger	the close breaks below the previous 60-bar low; the previous 60 bars' realised vol was $\leq 0.6 \times$ its time-of-day norm; the 5-minute move ≥ 1.0 sigma units
Parameters	{"N": 60, "cth": 0.6, "mode": "continuation", "strong": 1.0}
Signal timing	evaluated at the close of each 1-minute bar
Entry order	market order at the open of the next 1-minute bar (same session and contract)
Entry window	09:30–15:30 New York; one position at a time; no pyramiding
Initial stop	$1.0 \times \text{ATRh}$ from entry ($\text{ATRh} = \text{ATR}(14)$ of completed 15-minute bars); stop order, gaps through the stop fill at the bar open; rounded away from entry to the tick
Target	none (100% of the position runs)
Break-even / trailing / partials	none / none / none
Time exit	flat at the close of the last bar before 16:00 New York
Same-bar ambiguity	pessimistic (stop assumed first)
Costs (baseline)	\$14 per round trip = \$4 commission + 1 tick slippage per side; also scored at \$0 / \$9 / \$26
Cost file version	costs_v2_approved_2026-09-21#a27ed5df0c04

Statistical evidence trail

Stage	Test / statistic	Result	Verdict
Discovery 2010–18	label (trades ≥ 30 , net > 0 , $t \geq 1$)	742 trades, PF 1.17, t 1.36	PRIMARY
Discovery	parameter robustness (share of grid neighbours positive)	1.00	
Discovery	sub-periods positive (of 3) / years positive (of 9)	3 / 5	
Confirmation 2019–22	t-stat of mean trade	2.58	
Confirmation	null p (zero-drift worlds) / (drift worlds) / max	0.020 / 0.020 / 0.020	$p \leq 0.05$
Confirmation	family-wise p (max-t in same frequency bucket)	0.158	not significant
Confirmation	parameter support / management support	0.50 / 0.77	
Confirmation	years positive (of 4); stress net	4; \$54,634	
Confirmation	evidence tier (preregistered)		STRONG
Behaviour cluster	cluster at p 0.50 (members / entry rules)	#3 (13 / 1)	STRONG
Holdout 2023–25	t-stat; z vs confirmation mean trade	−0.19; −1.78	
Holdout	null p (100 holdout null worlds)	0.446	
Holdout	classification (preregistered)		FAILED
2026 audit	t-stat; net	1.66; \$26,378	SUPPORTED

Null worlds randomly flip the sign of each minute's moves (keeping volatility, seasonality, gaps and the NQ-ES relation) so any directional edge is destroyed; p is the share of such worlds that match the real t-stat. The family-wise p corrects for having searched many candidates of similar frequency.

H06-A (#6, long + short): performance



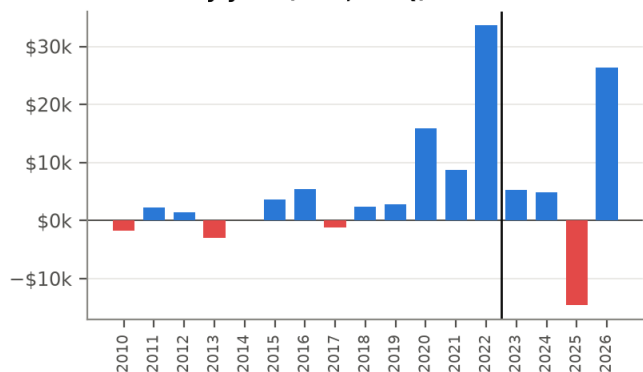
H06-A (#6, long + short): full statistics by period

Metric	Discovery 2010-18	Confirm. 2019-22	Holdout 2023-25	2026 Jan-Aug	2024-25	Post-freeze 2023-26	Full 2010-26
Trades	742	531	331	53	241	384	1,657
Trades / year	87	133	110	86	121	106	102
Long share	50%	55%	55%	55%	55%	55%	53%
Net P&L (baseline \$14)	\$9,162	\$61,006	-\$4,424	\$26,378	-\$9,669	\$21,954	\$92,122
Gross P&L (no costs)	\$19,550	\$68,440	\$210	\$27,120	-\$6,295	\$27,330	\$115,320
Net P&L, moderate \$9	\$12,872	\$63,661	-\$2,769	\$26,643	-\$8,464	\$23,874	\$100,407
Net P&L, stress \$26	\$258	\$54,634	-\$8,396	\$25,742	-\$12,561	\$17,346	\$72,238
Profit factor (net)	1.17	1.43	0.97	2.11	0.91	1.14	1.26
Profit factor (gross)	1.40	1.50	1.00	2.16	0.94	1.17	1.34
Profit factor (stress)	1.00	1.38	0.94	2.06	0.89	1.11	1.19
Win rate	32.6%	33.5%	29.0%	43.4%	27.8%	31.0%	32.5%
Average trade	\$12	\$115	-\$13	\$498	-\$40	\$57	\$56
Average winner	\$267	\$1,134	\$1,389	\$2,183	\$1,544	\$1,543	\$835
Average loser	-\$111	-\$399	-\$586	-\$795	-\$650	-\$610	-\$320
Payoff ratio (avg W / avg L)	2.41	2.84	2.37	2.75	2.38	2.53	2.61
Expectancy (net R / trade)	+0.019	+0.370	+0.097	+0.752	+0.087	+0.188	+0.171
Total net R	+14.5	+196.3	+32.2	+39.9	+21.0	+72.1	+282.8
Median trade (R)	-1.10	-1.02	-1.02	-0.16	-1.02	-1.02	-1.03
t-stat of mean trade (USD)	1.36	2.58	-0.19	1.66	-0.44	0.76	2.43
t-stat of mean trade (R)	0.25	2.90	0.63	1.80	0.46	1.30	2.68
Sharpe, daily, annualised (USD)	0.46	1.31	-0.10	2.04	-0.29	0.39	0.59
Sharpe, daily, annualised (R)	0.08	1.44	0.36	2.21	0.32	0.67	0.66
Sortino (USD)	0.92	3.14	-0.19	6.18	-0.55	0.81	1.31
Max drawdown (USD)	\$3,457	\$9,376	\$35,340	\$8,832	\$35,340	\$35,340	\$35,340
Max drawdown (R)	66.5	18.7	43.1	8.9	43.1	43.1	66.5
Longest drawdown (calendar days)	1,090	244	306	91	306	463	1,090
Calmar (annual net / max DD)	0.31	1.63	-0.04	4.85	-0.14	0.17	0.16
Longest losing streak (trades)	17	17	12	8	12	12	17
Longest winning streak	5	5	5	5	5	5	5
Best trade	\$1,621	\$5,646	\$10,061	\$10,896	\$10,061	\$10,896	\$10,896
Worst trade	-\$464	-\$1,239	-\$1,964	-\$1,449	-\$1,964	-\$1,964	-\$1,964
Best day / worst day	\$1,621 / -\$912	\$5,646 / -\$2,292	\$10,061 / -\$3,162	\$10,896 / -\$2,343	\$10,061 / -\$3,162	\$10,896 / -\$3,162	\$10,896 / -\$3,162
Top-10 trades as % of net	139%	80%	-	149%	-	302%	75%
Positive months	47% of 98	54% of 48	39% of 36	62% of 8	38% of 24	43% of 44	48% of 190
Average hold (minutes)	100	103	86	123	88	91	99
Avg initial risk (NQ points)	5.5	20.6	30.9	51.0	34.3	33.7	16.9
Avg initial risk, 1 NQ (USD) = 1R	\$110	\$413	\$618	\$1,020	\$685	\$673	\$338
Average NQ price at entry	4,296	11,082	19,585	27,188	21,451	20,634	10,257

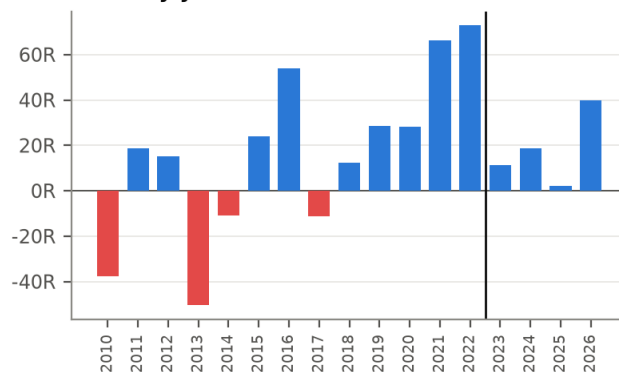
Sharpe and Sortino use every business day in the period (zero on no-trade days) and are scale-free. The strategy is always flat overnight.

H06-A (#6, long + short): calendar

Net P&L by year (USD, 1 NQ)



Net R by year

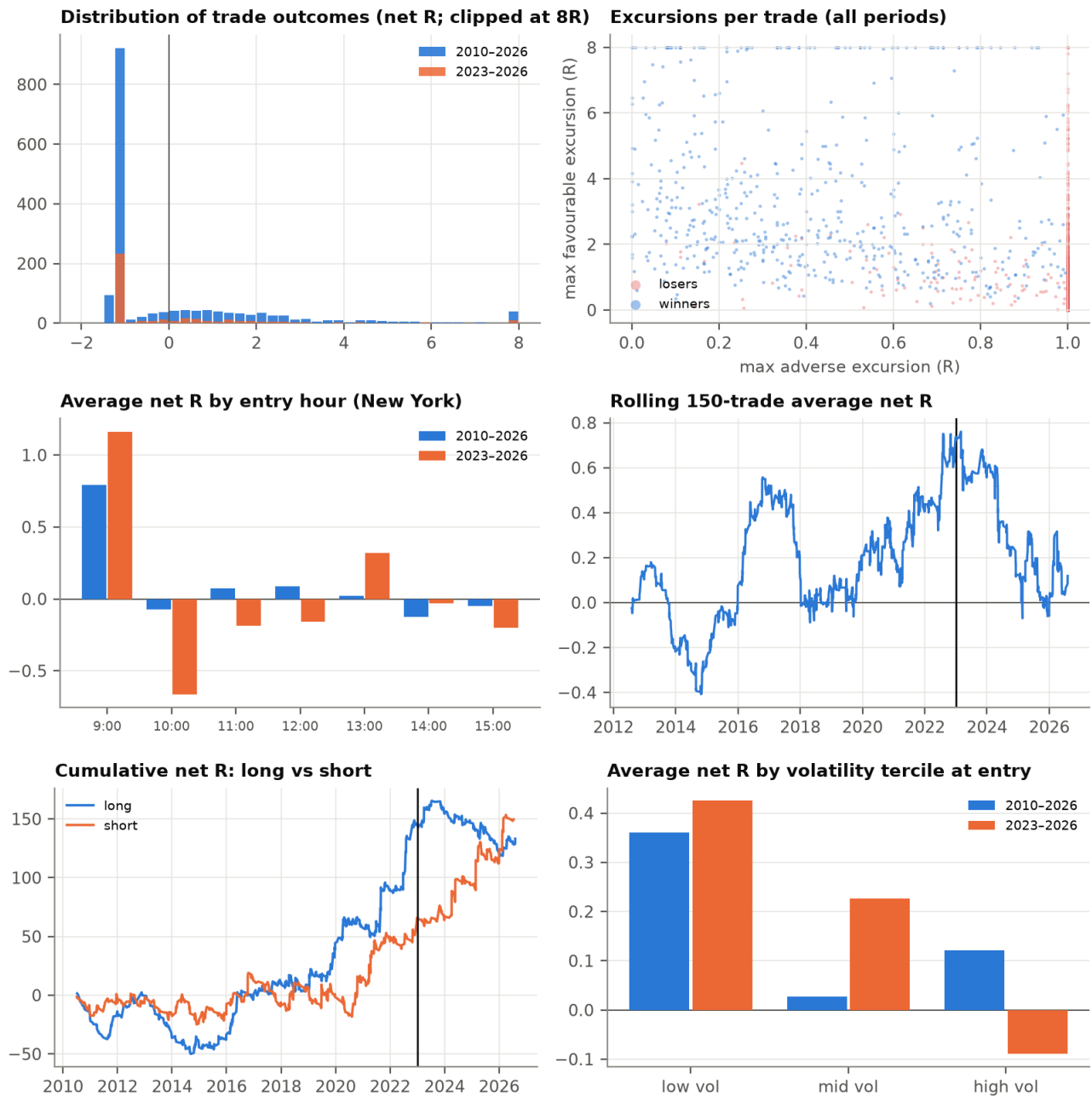


Year	Period	N	Net P&L	Gross	Stress	PF	Win %	Avg trade	Net R	Avg R	Max DD	1R (USD)
2010	Disc.	45	-\$1,775	-\$1,145	-\$2,315	0.25	16%	-\$39	-37.8	-0.840	\$1,864	\$54
2011	Disc.	70	\$2,290	\$3,270	\$1,450	1.70	36%	\$33	+18.8	+0.268	\$931	\$81
2012	Disc.	67	\$1,442	\$2,380	\$638	1.41	36%	\$22	+15.3	+0.228	\$1,061	\$80
2013	Disc.	85	-\$3,020	-\$1,830	-\$4,040	0.42	21%	-\$36	-50.4	-0.592	\$3,261	\$69
2014	Disc.	101	-\$9	\$1,405	-\$1,221	1.00	35%	-\$0	-10.7	-0.106	\$1,977	\$91
2015	Disc.	91	\$3,636	\$4,910	\$2,544	1.63	43%	\$40	+24.1	+0.265	\$829	\$110
2016	Disc.	79	\$5,384	\$6,490	\$4,436	1.96	41%	\$68	+54.1	+0.684	\$997	\$121
2017	Disc.	98	-\$1,207	\$165	-\$2,383	0.84	34%	-\$12	-11.3	-0.115	\$1,960	\$110
2018	Disc.	106	\$2,421	\$3,905	\$1,149	1.15	27%	\$23	+12.4	+0.117	\$2,594	\$214
2019	Conf.	153	\$2,858	\$5,000	\$1,022	1.13	33%	\$19	+28.6	+0.187	\$3,177	\$206
2020	Conf.	149	\$15,849	\$17,935	\$14,061	1.35	34%	\$106	+28.3	+0.190	\$6,368	\$491
2021	Conf.	139	\$8,674	\$10,620	\$7,006	1.21	32%	\$62	+66.4	+0.478	\$5,461	\$419
2022	Conf.	90	\$33,625	\$34,885	\$32,545	2.00	37%	\$374	+72.9	+0.810	\$7,065	\$625
2023	Holdout	90	\$5,245	\$6,505	\$4,165	1.21	32%	\$58	+11.3	+0.125	\$8,679	\$437
2024	Holdout	90	\$4,880	\$6,140	\$3,800	1.14	29%	\$54	+18.7	+0.208	\$7,328	\$557
2025	Holdout	151	-\$14,549	-\$12,435	-\$16,361	0.82	27%	-\$96	+2.2	+0.015	\$35,340	\$761
2026	Audit	53	\$26,378	\$27,120	\$25,742	2.11	43%	\$498	+39.9	+0.752	\$8,832	\$1,020

Monthly net R, 2019-2026 (blue = gain, red = loss; blank = no data)

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
2019	+4.7	-5.3	-2.2	-1.6	-2.1	+11.0	-0.1	-4.0	+8.8	+8.1	+3.6	+7.7
2020	-0.3	+2.1	-0.8	+10.7	-9.8	+2.0	-1.8	+10.5	+5.3	+18.2	-6.0	-1.8
2021	-1.0	-1.9	+9.9	+11.7	+0.6	+11.7	-3.8	+31.5	+4.8	+4.6	+4.7	-6.4
2022	-1.1	+0.5	-2.3	-1.0	-2.2	+14.9	+27.1	+7.9	-2.1	+10.2	+6.0	+14.9
2023	-1.4	-2.9	+13.3	+2.1	-1.2	+2.2	+8.7	-1.0	-3.1	+13.5	-13.2	-5.6
2024	+0.8	-5.7	-1.3	+19.0	+16.6	+0.4	-4.2	-0.2	-5.9	+2.6	+0.0	-3.5
2025	-2.7	+18.3	+5.4	+13.4	-12.7	-3.0	+1.3	-5.3	-7.3	-4.3	+1.2	-2.1
2026	-0.5	+33.2	+1.9	+5.3	-1.0	-2.5	-1.5	+5.0				

H06-A (#6, long + short): trade anatomy



Volatility tercile = initial stop distance as a share of price, ranked within each calendar year.

H06-A (#6, long + short): breakdowns (all periods vs post-freeze 2023-26)

Direction

Direction	Trades (all)	Avg R (all)	PF (all)	Net (all)	Trades 23-26	Avg R 23-26	PF 23-26	Net 23-26
Long	876	+0.152	1.25	\$46,586	212	−0.055	0.96	−\$3,718
Short	781	+0.192	1.27	\$45,536	172	+0.487	1.35	\$25,672

Exit reason

Exit	Trades (all)	Avg R (all)	PF (all)	Net (all)	Trades 23-26	Avg R 23-26	PF 23-26	Net 23-26
16:00 flat	642	+2.183	28.35	\$434,297	150	+2.084	17.81	\$173,290
Stop-loss hit	1015	−1.102	0.00	−\$342,175	234	−1.028	0.00	−\$151,336

Entry hour (New York)

Entry hour	Trades (all)	Avg R (all)	PF (all)	Net (all)	Trades 23-26	Avg R 23-26	PF 23-26	Net 23-26
09:00-09:59	373	+0.793	2.52	\$102,563	87	+1.163	2.93	\$51,617
10:00-10:59	147	−0.074	0.74	−\$10,223	33	−0.662	0.27	−\$11,912
11:00-11:59	204	+0.075	1.06	\$2,844	46	−0.184	0.82	−\$4,564
12:00-12:59	241	+0.088	0.95	−\$2,754	50	−0.160	0.66	−\$7,950
13:00-13:59	248	+0.022	1.21	\$11,888	59	+0.320	1.26	\$7,249
14:00-14:59	295	−0.125	0.81	−\$11,465	72	−0.031	0.78	−\$5,973
15:00-15:59	149	−0.049	0.97	−\$731	37	−0.199	0.57	−\$6,513

Weekday

Weekday	Trades (all)	Avg R (all)	PF (all)	Net (all)	Trades 23-26	Avg R 23-26	PF 23-26	Net 23-26
Mon	424	+0.221	1.58	\$40,779	90	−0.022	1.12	\$3,485
Tue	340	+0.199	1.01	\$460	91	−0.277	0.62	−\$16,799
Wed	318	+0.202	1.23	\$19,368	90	+0.256	1.14	\$5,890
Thu	250	+0.187	1.44	\$25,175	54	+0.996	1.99	\$23,889
Fri	325	+0.032	1.11	\$6,340	59	+0.381	1.27	\$5,489

H06-A (#6, long + short): robustness

Parameter neighbourhood. Every grid point of the family with the same stop and exit, both directions separately. If the frozen point is a lucky outlier, its neighbours will look much worse. Rerun post hoc for this report; purely descriptive, and it does not re-select anything.

Entry parameters	Dir.	Disc. n / PF	Conf. n / PF	Holdout n / PF	Holdout net	2026 n / PF	2026 net
N=30, cth=0.6, strong=1.0	long	708 / 1.03	429 / 1.18	304 / 0.79	-\$28,131	53 / 1.10	\$3,023
N=30, cth=0.6, strong=1.0	short	741 / 0.85	463 / 1.09	287 / 0.93	-\$9,038	71 / 1.55	\$20,461
N=30, cth=0.6, strong=1.0	both	1213 / 1.03	752 / 1.18	501 / 0.88	-\$26,654	100 / 1.58	\$32,055
N=30, cth=0.6, strong=None	long	1113 / 1.08	676 / 1.25	487 / 0.88	-\$23,273	80 / 0.96	-\$2,030
N=30, cth=0.6, strong=None	short	1170 / 0.80	701 / 0.90	463 / 0.92	-\$16,347	91 / 1.42	\$20,816
N=30, cth=0.6, strong=None	both	1779 / 1.04	1053 / 1.17	743 / 0.91	-\$28,432	126 / 1.42	\$30,991
N=30, cth=0.75, strong=1.0	long	1747 / 0.98	906 / 1.30	657 / 0.98	-\$4,433	117 / 0.91	-\$7,548
N=30, cth=0.75, strong=1.0	short	1877 / 0.91	977 / 1.04	660 / 1.08	\$22,350	143 / 1.05	\$5,063
N=30, cth=0.75, strong=1.0	both	2878 / 1.00	1470 / 1.23	1057 / 1.07	\$29,712	191 / 1.10	\$12,331
N=30, cth=0.75, strong=None	long	2244 / 0.99	1171 / 1.24	868 / 1.00	\$778	140 / 1.01	\$1,180
N=30, cth=0.75, strong=None	short	2391 / 0.89	1248 / 0.98	837 / 1.06	\$23,022	172 / 1.01	\$1,362
N=30, cth=0.75, strong=None	both	3524 / 1.00	1763 / 1.22	1292 / 1.06	\$34,857	220 / 1.15	\$21,535
N=30, cth=None, strong=1.0	long	4829 / 0.94	2261 / 1.07	1724 / 1.04	\$38,369	345 / 0.95	-\$16,165
N=30, cth=None, strong=1.0	short	4963 / 0.94	2272 / 1.04	1741 / 0.95	-\$50,169	336 / 1.05	\$13,716
N=30, cth=None, strong=1.0	both	7230 / 0.96	3288 / 1.13	2542 / 1.04	\$52,877	492 / 1.03	\$14,822
N=30, cth=None, strong=None	long	5358 / 0.94	2502 / 1.05	1931 / 1.04	\$38,551	374 / 0.94	-\$21,501
N=30, cth=None, strong=None	short	5517 / 0.93	2544 / 1.00	1918 / 0.94	-\$60,812	370 / 1.01	\$4,570
N=30, cth=None, strong=None	both	7933 / 0.95	3593 / 1.10	2788 / 1.03	\$47,893	534 / 0.99	-\$5,941
N=60, cth=0.6, strong=1.0	long	390 / 1.20	302 / 1.56	190 / 0.83	-\$13,655	30 / 1.74	\$9,825
N=60, cth=0.6, strong=1.0	short	420 / 1.01	281 / 1.19	162 / 1.15	\$10,322	30 / 2.14	\$14,705
N=60, cth=0.6, strong=1.0 ◀ frozen	both	742 / 1.17	531 / 1.43	331 / 0.97	-\$4,424	53 / 2.11	\$26,378
N=60, cth=0.6, strong=None	long	658 / 1.10	464 / 1.57	330 / 0.87	-\$16,840	52 / 1.11	\$3,022
N=60, cth=0.6, strong=None	short	628 / 0.88	410 / 0.98	254 / 1.11	\$10,599	47 / 1.38	\$9,452
N=60, cth=0.6, strong=None	both	1138 / 1.00	753 / 1.37	517 / 0.98	-\$3,888	84 / 1.47	\$19,484
N=60, cth=0.75, strong=1.0	long	1239 / 1.03	693 / 1.29	486 / 0.94	-\$11,664	84 / 0.92	-\$3,946
N=60, cth=0.75, strong=1.0	short	1243 / 1.08	672 / 1.07	439 / 1.06	\$12,129	94 / 0.90	-\$7,361
N=60, cth=0.75, strong=1.0	both	2198 / 1.09	1182 / 1.23	821 / 0.99	-\$3,409	152 / 0.98	-\$2,313
N=60, cth=0.75, strong=None	long	1624 / 0.98	888 / 1.24	640 / 0.96	-\$11,160	108 / 0.89	-\$7,452
N=60, cth=0.75, strong=None	short	1556 / 0.98	853 / 0.93	562 / 1.05	\$11,862	114 / 0.93	-\$5,491
N=60, cth=0.75, strong=None	both	2753 / 1.00	1446 / 1.17	1021 / 1.02	\$8,261	188 / 0.91	-\$11,837
N=60, cth=None, strong=1.0	long	3974 / 0.94	1865 / 1.09	1423 / 1.01	\$7,773	283 / 1.02	\$6,118
N=60, cth=None, strong=1.0	short	4050 / 0.98	1883 / 1.02	1409 / 0.99	-\$6,426	289 / 0.98	-\$5,316
N=60, cth=None, strong=1.0	both	6585 / 0.96	3047 / 1.11	2329 / 1.04	\$52,064	456 / 1.02	\$7,886
N=60, cth=None, strong=None	long	4452 / 0.94	2079 / 1.05	1589 / 1.03	\$20,904	315 / 0.96	-\$10,525
N=60, cth=None, strong=None	short	4464 / 0.97	2081 / 0.97	1539 / 0.99	-\$10,526	321 / 0.95	-\$14,674
N=60, cth=None, strong=None	both	7260 / 0.96	3315 / 1.08	2527 / 1.05	\$67,347	509 / 0.95	-\$22,551

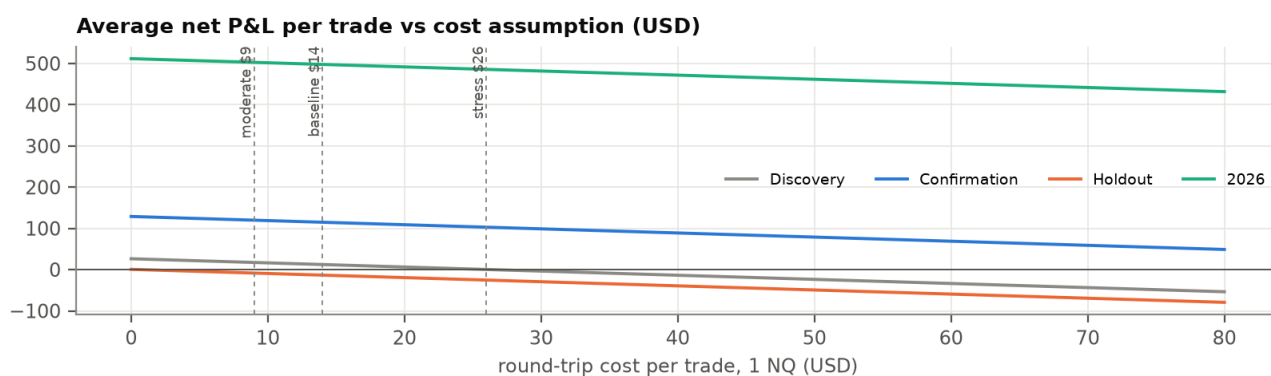
Neighbours with ≥ 20 trades and positive net: Discovery 14/36; Confirmation 31/36; Holdout 18/36; 2026 21/36

Stop / exit management. The frozen entry under every management variant from the campaign library (discovery stages A and B).

Stop / exit management	Disc. PF	Conf. PF	Holdout n / PF	Holdout net	2026 PF	2026 net
1×ATR stop; single target at +1R	0.94	1.04	401 / 0.72	-\$38,239	1.22	\$5,314
1×ATR stop; single target at +1.5R	1.04	1.07	380 / 0.73	-\$39,460	1.19	\$5,120
1×ATR stop; single target at +2R	1.11	1.11	368 / 0.79	-\$30,777	1.38	\$9,517
1×ATR stop; single target at +3R	1.16	1.17	355 / 0.88	-\$17,685	1.50	\$12,682
1×ATR stop; single target at +4R	1.20	1.27	349 / 0.89	-\$15,841	1.57	\$14,145
1×ATR stop; no target, flat 16:00 ◀ frozen	1.17	1.43	331 / 0.97	-\$4,424	2.11	\$26,378
1×ATR stop; no target, exit after 60 min	1.13	1.22	382 / 0.95	-\$6,318	2.16	\$22,574
1.5×ATR stop; single target at +1R	1.09	1.12	361 / 0.85	-\$22,029	1.12	\$3,418
1.5×ATR stop; single target at +2R	1.18	1.26	328 / 1.01	\$1,068	1.45	\$12,165

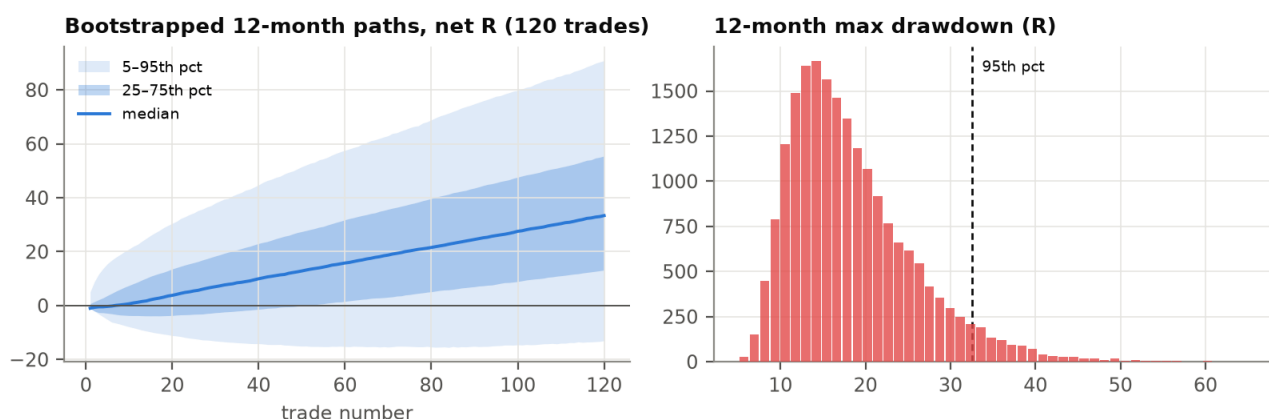
Stop / exit management	Disc. PF	Conf. PF	Holdout n / PF	Holdout net	2026 PF	2026 net
1.5×ATR stop; no target, flat 16:00	1.18	1.57	307 / 1.05	\$7,117	1.83	\$22,322
5-min swing stop; single target at +1R	1.08	1.01	335 / 0.98	−\$3,180	1.39	\$11,118
5-min swing stop; single target at +2R	1.23	1.16	308 / 0.96	−\$6,102	1.67	\$17,949
5-min swing stop; no target, flat 16:00	1.15	1.26	293 / 1.02	\$2,383	1.90	\$24,272
1×ATR stop; stop to entry at +0.5R, target +2R	0.79	1.00	397 / 0.46	−\$52,903	1.65	\$8,821
1×ATR stop; stop to entry at +0.5R, no target	0.83	1.11	378 / 0.52	−\$43,512	1.76	\$10,425
1×ATR stop; stop to entry at +1R, target +2R	1.00	1.04	374 / 0.72	−\$36,051	1.30	\$5,952
1×ATR stop; stop to entry at +1R, no target	1.04	1.35	345 / 0.71	−\$34,345	1.51	\$10,295
1×ATR stop; stop to entry at +1.5R, target +2R	1.11	1.09	370 / 0.75	−\$35,425	1.26	\$6,282
1×ATR stop; stop to entry at +1.5R, no target	1.18	1.41	338 / 0.88	−\$15,787	1.77	\$18,715
1×ATR stop; no target, 1.5×ATR trailing stop	1.09	1.14	360 / 0.85	−\$18,185	1.34	\$7,339
1×ATR stop; no target, 2×ATR trailing stop	1.21	1.24	348 / 0.89	−\$14,872	1.09	\$2,137
1×ATR stop; no target, 3×ATR trailing stop	1.20	1.30	336 / 0.93	−\$9,389	1.76	\$18,883
1×ATR stop; no target, trail beyond last 5-min swing	1.10	1.21	356 / 0.94	−\$8,354	1.76	\$18,648
1×ATR stop; 50% at +1R, 50% at +2R	1.01	1.07	368 / 0.73	−\$34,704	1.44	\$8,732
1×ATR stop; 50% at +1R, 50% at +3R	1.05	1.12	355 / 0.79	−\$26,005	1.51	\$10,152
1×ATR stop; 50% at +1R, rest trails 2×ATR	1.06	1.15	346 / 0.80	−\$24,264	1.27	\$5,274
1×ATR stop; 25% at +1R, 75% runs to 16:00	1.12	1.38	331 / 0.91	−\$11,573	2.03	\$21,620
1×ATR stop; 50% at +1R then stop to entry, rest +3R	0.97	1.11	361 / 0.75	−\$30,802	1.39	\$7,807
1×ATR stop; stop to entry at +1R, 2×ATR trail	1.06	1.18	354 / 0.79	−\$24,531	1.05	\$1,092

Cost sensitivity. The line shows what the average trade would be at any round-trip cost.



Break-even round-trip cost per contract (gross average trade): Discovery \$26; Confirmation \$129; Holdout \$1; 2026 \$512. The baseline assumption is \$14.

H06-A (#6, long + short): risk, sizing and forward expectations



Bootstrap of 915 trades from 2019–Aug 2026 (all out-of-sample relative to discovery), 120 trades per simulated year, 20,000 paths, trades drawn independently. This ignores regime persistence, so real drawdowns cluster more than shown.

Forward 12-month statistic (bootstrap)	In R	1 NQ at 2026 risk (1R ≈ \$1,020)	1 MNQ (1R ≈ \$102)
Median 12-month result	+33.4	\$34,036	\$3,404
5th percentile (bad year)	−13.0	−\$13,310	−\$1,331
95th percentile (good year)	+90.8	\$92,640	\$9,264
Probability of a losing year	13%		
Median max drawdown	16.8	\$17,188	\$1,719
95th pct max drawdown	32.6	\$33,233	\$3,323
99th pct max drawdown	41.1	\$41,905	\$4,191
Historical max drawdown 2010–26	66.5		

Suggested capital rule of thumb (not a margin requirement): hold at least the 99th-percentile 12-month drawdown plus exchange margin per unit traded. Dollar outcomes on one full NQ contract are far larger today than in most of the backtest; micro contracts (MNQ, 1/10 size) allow risk to be held near-constant per trade.

Portfolio context (25-strategy frozen cohort, 1 contract each)

Period	Corr. with rest of cohort (daily)	Median pairwise corr.	Most correlated members	Share of cohort P&L
holdout	0.22	0.03	#18 H06 long (0.43), #12 H43 both (0.18), #19 H41 both (0.16), #9 H08 short (0.16)	-1%
2026	0.34	0.13	#18 H06 long (0.42), #21 H25 short (0.39), #22 H47 long (0.35), #19 H41 both (0.29)	26%

H06-B (#18, long only): specification

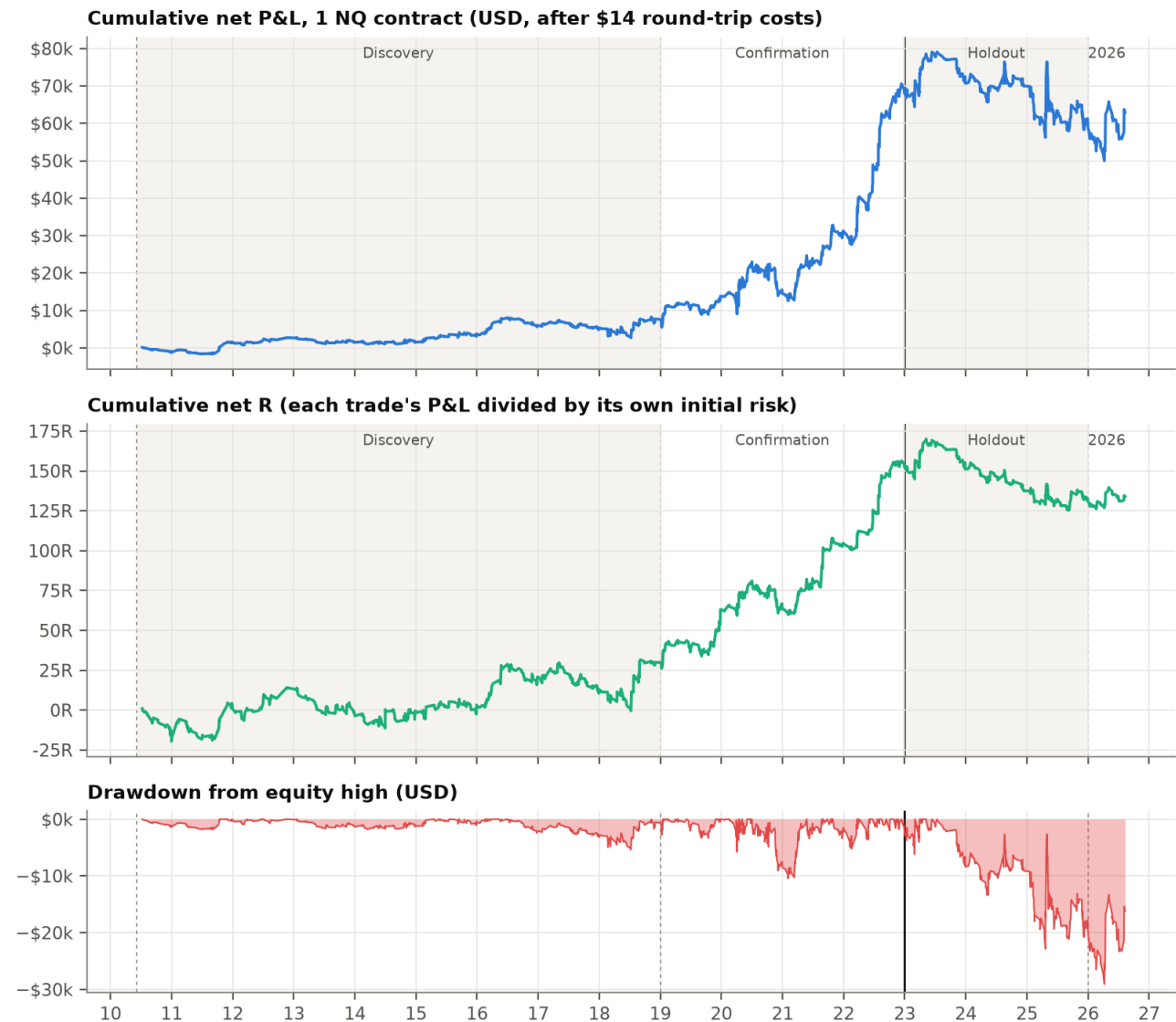
Field	Frozen value
Strategy ID	C39c52e02a3c54412
Entry rule ID	E866a4380cd32ac16
Family / version	v3_h06_compression_expansion (v3.0)
Instrument	NQ (E-mini Nasdaq-100 futures, \$20 per point), front contract, 1-minute bars
Direction	long
Mode	continuation
Long trigger	the close breaks above the previous 60-bar high; the previous 60 bars' realised vol was $\leq 0.6 \times$ its time-of-day norm
Short trigger	not traded
Parameters	{"N": 60, "cth": 0.6, "mode": "continuation", "strong": null}
Signal timing	evaluated at the close of each 1-minute bar
Entry order	market order at the open of the next 1-minute bar (same session and contract)
Entry window	09:30–15:30 New York; one position at a time; no pyramiding
Initial stop	$1.5 \times \text{ATRh}$ from entry ($\text{ATRh} = \text{ATR}(14)$ of completed 15-minute bars); stop order, gaps through the stop fill at the bar open; rounded away from entry to the tick
Target	none (100% of the position runs)
Break-even / trailing / partials	none / none / none
Time exit	flat at the close of the last bar before 16:00 New York
Same-bar ambiguity	pessimistic (stop assumed first)
Costs (baseline)	\$14 per round trip = \$4 commission + 1 tick slippage per side; also scored at \$0 / \$9 / \$26
Cost file version	costs_v2_approved_2026-09-21#a27ed5df0c04

Statistical evidence trail

Stage	Test / statistic	Result	Verdict
Discovery 2010–18	label (trades ≥ 30 , net > 0 , $t \geq 1$)	618 trades, PF 1.15, t 1.15	PRIMARY
Discovery	parameter robustness (share of grid neighbours positive)	1.00	
Discovery	sub-periods positive (of 3) / years positive (of 9)	3 / 6	
Confirmation 2019–22	t-stat of mean trade	2.79	
Confirmation	null p (zero-drift worlds) / (drift worlds) / max	0.020 / 0.020 / 0.020	$p \leq 0.05$
Confirmation	family-wise p (max-t in same frequency bucket)	0.099	significant at 10%
Confirmation	parameter support / management support	0.00 / 0.31	
Confirmation	years positive (of 4); stress net	4; \$55,070	
Confirmation	evidence tier (preregistered)		PROMISING
Behaviour cluster	cluster at p 0.50 (members / entry rules)	#2 (19 / 2)	PROMISING
Holdout 2023–25	t-stat; z vs confirmation mean trade	−0.44; −2.46	
Holdout	null p (100 holdout null worlds)	0.644	
Holdout	classification (preregistered)		FAILED
2026 audit	t-stat; net	0.32; \$3,947	WEAKER_THAN_EXPECTED

Null worlds randomly flip the sign of each minute's moves (keeping volatility, seasonality, gaps and the NQ-ES relation) so any directional edge is destroyed; p is the share of such worlds that match the real t-stat. The family-wise p corrects for having searched many candidates of similar frequency.

H06-B (#18, long only): performance



Shaded bands mark the research stages; the solid black line is the start of data never seen before the freeze (1 Jan 2023).

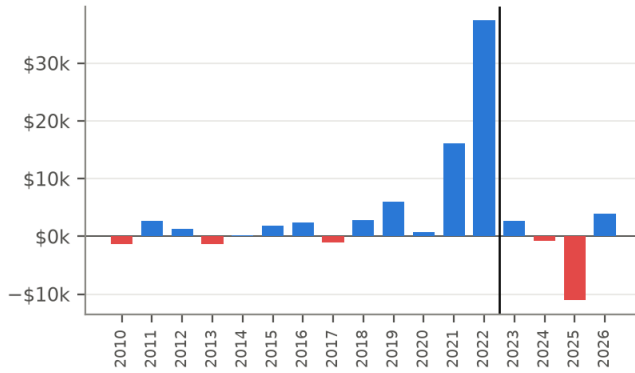
H06-B (#18, long only): full statistics by period

Metric	Discovery 2010-18	Confirm. 2019-22	Holdout 2023-25	2026 Jan-Aug	2024-25	Post-freeze 2023-26	Full 2010-26
Trades	618	435	303	47	214	350	1,403
Trades / year	72	109	101	76	107	97	87
Long share	100%	100%	100%	100%	100%	100%	100%
Net P&L (baseline \$14)	\$7,733	\$60,290	-\$9,047	\$3,947	-\$11,736	-\$5,100	\$62,923
Gross P&L (no costs)	\$16,385	\$66,380	-\$4,805	\$4,605	-\$8,740	-\$200	\$82,565
Net P&L, moderate \$9	\$10,823	\$62,465	-\$7,532	\$4,182	-\$10,666	-\$3,350	\$69,938
Net P&L, stress \$26	\$317	\$55,070	-\$12,683	\$3,383	-\$14,304	-\$9,300	\$46,087
Profit factor (net)	1.15	1.48	0.93	1.13	0.89	0.97	1.18
Profit factor (gross)	1.35	1.54	0.96	1.16	0.91	1.00	1.25
Profit factor (stress)	1.01	1.43	0.91	1.11	0.86	0.94	1.13
Win rate	39.6%	46.0%	35.6%	38.3%	35.5%	36.0%	40.7%
Average trade	\$13	\$139	-\$30	\$84	-\$55	-\$15	\$45
Average winner	\$243	\$930	\$1,165	\$1,861	\$1,211	\$1,265	\$709
Average loser	-\$139	-\$535	-\$692	-\$1,019	-\$752	-\$734	-\$411
Payoff ratio (avg W / avg L)	1.75	1.74	1.68	1.83	1.61	1.72	1.73
Expectancy (net R / trade)	+0.049	+0.282	-0.072	+0.068	-0.095	-0.053	+0.096
Total net R	+30.0	+122.5	-21.7	+3.2	-20.3	-18.5	+134.0
Median trade (R)	-0.56	-0.31	-0.71	-0.36	-0.50	-0.57	-0.52
t-stat of mean trade (USD)	1.15	2.79	-0.44	0.32	-0.63	-0.21	1.90
t-stat of mean trade (R)	0.68	2.94	-0.91	0.34	-1.12	-0.72	2.03
Sharpe, daily, annualised (USD)	0.39	1.40	-0.24	0.40	-0.43	-0.11	0.47
Sharpe, daily, annualised (R)	0.23	1.45	-0.51	0.42	-0.76	-0.37	0.50
Sortino (USD)	0.75	2.86	-0.39	0.76	-0.69	-0.18	0.85
Max drawdown (USD)	\$5,356	\$10,414	\$22,839	\$10,033	\$20,189	\$29,062	\$29,062
Max drawdown (R)	30.2	21.1	44.8	8.6	29.9	44.8	44.8
Longest drawdown (calendar days)	858	323	902	97	499	1,124	1,124
Calmar (annual net / max DD)	0.17	1.45	-0.13	0.64	-0.29	-0.05	0.13
Longest losing streak (trades)	9	9	9	4	9	9	9
Longest winning streak	7	7	6	7	6	7	7
Best trade	\$1,936	\$5,646	\$6,316	\$5,986	\$6,316	\$6,316	\$6,316
Worst trade	-\$894	-\$1,789	-\$2,369	-\$2,344	-\$2,369	-\$2,369	-\$2,369
Best day / worst day	\$1,936 / -\$1,072	\$5,132 / -\$2,352	\$6,316 / -\$3,673	\$5,986 / -\$2,344	\$6,316 / -\$3,673	\$6,316 / -\$3,673	\$6,316 / -\$3,673
Top-10 trades as % of net	168%	69%	-	743%	-	-	78%
Positive months	48% of 96	60% of 48	34% of 35	38% of 8	33% of 24	35% of 43	48% of 187
Average hold (minutes)	127	133	117	143	122	120	127
Avg initial risk (NQ points)	8.1	30.9	45.3	76.9	51.4	49.6	25.5
Avg initial risk, 1 NQ (USD) = 1R	\$161	\$617	\$907	\$1,538	\$1,027	\$992	\$510
Average NQ price at entry	4,312	11,129	19,267	27,771	21,341	20,409	10,441

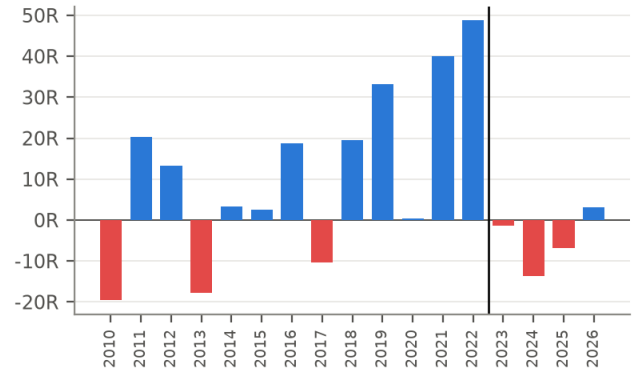
Sharpe and Sortino use every business day in the period (zero on no-trade days) and are scale-free. The strategy is always flat overnight.

H06-B (#18, long only): calendar

Net P&L by year (USD, 1 NQ)



Net R by year

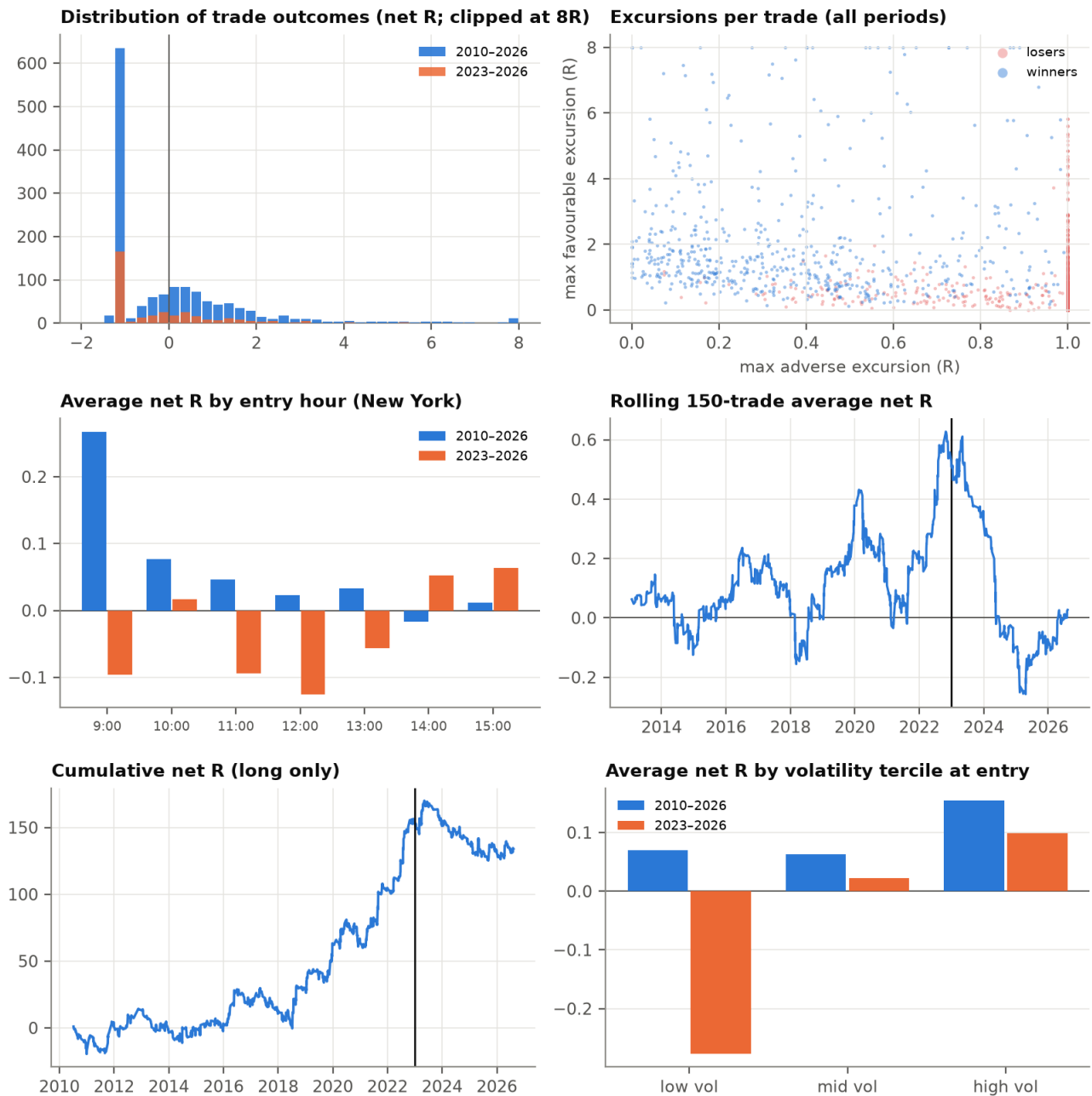


Year	Period	N	Net P&L	Gross	Stress	PF	Win %	Avg trade	Net R	Avg R	Max DD	1R (USD)
2010	Disc.	38	-\$1,287	-\$755	-\$1,743	0.41	29%	-\$34	-19.5	-0.514	\$1,413	\$77
2011	Disc.	59	\$2,649	\$3,475	\$1,941	1.85	47%	\$45	+20.4	+0.345	\$1,189	\$112
2012	Disc.	45	\$1,355	\$1,985	\$815	1.61	51%	\$30	+13.3	+0.295	\$747	\$112
2013	Disc.	65	-\$1,280	-\$370	-\$2,060	0.69	34%	-\$20	-17.7	-0.272	\$1,657	\$104
2014	Disc.	88	\$213	\$1,445	-\$843	1.04	41%	\$2	+3.2	+0.037	\$919	\$131
2015	Disc.	80	\$1,905	\$3,025	\$945	1.31	41%	\$24	+2.5	+0.032	\$1,067	\$169
2016	Disc.	73	\$2,398	\$3,420	\$1,522	1.40	44%	\$33	+18.8	+0.257	\$2,411	\$180
2017	Disc.	91	-\$1,099	\$175	-\$2,191	0.86	38%	-\$12	-10.4	-0.114	\$2,520	\$162
2018	Disc.	79	\$2,879	\$3,985	\$1,931	1.20	32%	\$36	+19.5	+0.247	\$2,921	\$322
2019	Conf.	115	\$6,075	\$7,685	\$4,695	1.39	50%	\$53	+33.2	+0.289	\$3,321	\$317
2020	Conf.	138	\$693	\$2,625	-\$963	1.01	41%	\$5	+0.4	+0.003	\$9,060	\$701
2021	Conf.	109	\$16,099	\$17,625	\$14,791	1.59	47%	\$148	+40.1	+0.368	\$3,419	\$603
2022	Conf.	73	\$37,423	\$38,445	\$36,547	2.28	49%	\$513	+48.8	+0.668	\$3,611	\$952
2023	Holdout	89	\$2,689	\$3,935	\$1,621	1.09	36%	\$30	-1.4	-0.016	\$8,376	\$618
2024	Holdout	84	-\$716	\$460	-\$1,724	0.98	32%	-\$9	-13.6	-0.161	\$7,449	\$776
2025	Holdout	130	-\$11,020	-\$9,200	-\$12,580	0.85	38%	-\$85	-6.7	-0.052	\$18,477	\$1,190
2026	Audit	47	\$3,947	\$4,605	\$3,383	1.13	38%	\$84	+3.2	+0.068	\$10,033	\$1,538

Monthly net R, 2019-2026 (blue = gain, red = loss; blank = no data)

Year	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
2019	+11.2	+0.7	-1.1	+2.0	-0.6	-5.8	+5.0	-5.3	+0.2	+5.4	+8.7	+12.9
2020	-0.6	+2.3	-2.5	+10.0	+1.1	+7.1	-2.3	-4.9	-3.2	+4.7	-10.0	-1.4
2021	-1.6	-0.6	+8.3	+5.1	+3.2	+3.5	-0.6	+20.9	-0.9	+6.4	-2.8	-0.8
2022	-0.6	-2.6	+10.9	+0.9	-0.1	+12.4	+10.0	+11.0	+1.9	+7.9	-2.8	-0.1
2023	-3.8	-3.7	+16.4	+6.8	-1.0	+0.7	-1.0	-2.0	-1.5		-7.8	-4.4
2024	+4.0	-3.4	-4.6	-2.1	+4.5	-2.1	-3.0	-0.3	-2.7	+1.4	-2.4	-2.8
2025	-0.3	-5.9	+0.3	+10.1	-10.8	+1.2	-4.0	-2.4	+11.4	+0.1	-1.9	-4.4
2026	-3.1	+3.3	-2.0	+8.2	-2.2	-0.8	-3.1	+2.9				

H06-B (#18, long only): trade anatomy



Volatility tercile = initial stop distance as a share of price, ranked within each calendar year.

H06-B (#18, long only): breakdowns (all periods vs post-freeze 2023-26)

Exit reason

Exit	Trades (all)	Avg R (all)	PF (all)	Net (all)	Trades 23-26	Avg R 23-26	PF 23-26	Net 23-26
16:00 flat	750	+1.116	10.37	\$365,865	185	+0.812	6.65	\$135,370
Stop-loss hit	653	−1.076	0.00	−\$302,942	165	−1.023	0.00	−\$140,470

Entry hour (New York)

Entry hour	Trades (all)	Avg R (all)	PF (all)	Net (all)	Trades 23-26	Avg R 23-26	PF 23-26	Net 23-26
09:00-09:59	401	+0.267	1.39	\$39,316	101	−0.097	1.08	\$3,546
10:00-10:59	135	+0.077	1.22	\$7,720	35	+0.017	1.43	\$5,590
11:00-11:59	183	+0.046	1.09	\$4,583	46	−0.094	0.95	−\$1,364
12:00-12:59	178	+0.023	0.85	−\$8,402	49	−0.125	0.71	−\$8,726
13:00-13:59	197	+0.033	1.25	\$12,117	50	−0.056	0.83	−\$4,160
14:00-14:59	220	−0.017	1.13	\$4,870	44	+0.053	1.00	−\$51
15:00-15:59	89	+0.012	1.21	\$2,719	25	+0.064	1.01	\$65

Weekday

Weekday	Trades (all)	Avg R (all)	PF (all)	Net (all)	Trades 23-26	Avg R 23-26	PF 23-26	Net 23-26
Mon	388	+0.248	1.61	\$48,043	102	−0.020	1.02	\$792
Tue	288	+0.126	1.06	\$4,568	76	−0.026	1.05	\$1,976
Wed	255	−0.101	0.85	−\$11,625	72	−0.171	0.76	−\$9,423
Thu	221	+0.026	1.10	\$5,306	47	−0.130	0.95	−\$1,123
Fri	251	+0.086	1.35	\$16,631	53	+0.075	1.14	\$2,678

H06-B (#18, long only): robustness

Parameter neighbourhood. Every grid point of the family with the same stop and exit, both directions separately. If the frozen point is a lucky outlier, its neighbours will look much worse. Rerun post hoc for this report; purely descriptive, and it does not re-select anything.

Entry parameters	Dir.	Disc. n / PF	Conf. n / PF	Holdout n / PF	Holdout net	2026 n / PF	2026 net
N=30, cth=0.6, strong=1.0	long	674 / 1.07	407 / 1.26	275 / 0.92	-\$11,090	50 / 0.98	-\$560
N=30, cth=0.6, strong=1.0	short	711 / 0.89	425 / 1.09	271 / 0.87	-\$19,854	66 / 1.58	\$20,871
N=30, cth=0.6, strong=1.0	both	1102 / 1.07	664 / 1.27	432 / 0.97	-\$6,043	93 / 1.40	\$24,258
N=30, cth=0.6, strong=None	long	1026 / 1.04	611 / 1.33	436 / 0.94	-\$12,629	74 / 0.87	-\$7,286
N=30, cth=0.6, strong=None	short	1069 / 0.82	621 / 0.94	416 / 0.89	-\$24,384	83 / 1.38	\$19,013
N=30, cth=0.6, strong=None	both	1542 / 1.00	880 / 1.28	627 / 0.94	-\$18,188	111 / 1.23	\$18,081
N=30, cth=0.75, strong=1.0	long	1585 / 1.02	810 / 1.37	596 / 1.00	\$1,386	109 / 0.80	-\$20,016
N=30, cth=0.75, strong=1.0	short	1716 / 0.94	874 / 1.04	595 / 1.04	\$13,060	128 / 1.05	\$5,388
N=30, cth=0.75, strong=1.0	both	2479 / 1.01	1229 / 1.32	890 / 1.13	\$62,410	166 / 1.01	\$861
N=30, cth=0.75, strong=None	long	1974 / 1.00	1016 / 1.31	747 / 1.07	\$25,077	128 / 0.94	-\$6,717
N=30, cth=0.75, strong=None	short	2120 / 0.91	1082 / 0.99	742 / 1.03	\$12,352	154 / 0.97	-\$3,606
N=30, cth=0.75, strong=None	both	2945 / 1.00	1415 / 1.30	1063 / 1.13	\$69,953	196 / 0.98	-\$3,334
N=30, cth=None, strong=1.0	long	4083 / 0.95	1915 / 1.04	1447 / 1.06	\$55,752	276 / 1.04	\$13,146
N=30, cth=None, strong=1.0	short	4153 / 0.93	1881 / 1.03	1433 / 0.96	-\$43,907	296 / 0.93	-\$21,924
N=30, cth=None, strong=1.0	both	5786 / 0.96	2646 / 1.07	1981 / 1.09	\$118,886	389 / 1.03	\$12,129
N=30, cth=None, strong=None	long	4484 / 0.95	2082 / 1.03	1598 / 1.04	\$39,438	296 / 1.03	\$8,476
N=30, cth=None, strong=None	short	4537 / 0.93	2068 / 0.99	1562 / 0.95	-\$54,643	322 / 0.90	-\$34,883
N=30, cth=None, strong=None	both	6323 / 0.93	2848 / 1.05	2128 / 1.09	\$118,348	421 / 0.99	-\$6,244
N=60, cth=0.6, strong=1.0	long	382 / 1.22	287 / 1.61	181 / 0.90	-\$8,924	30 / 1.42	\$6,960
N=60, cth=0.6, strong=1.0	short	411 / 0.98	267 / 1.34	158 / 1.13	\$10,188	29 / 1.95	\$13,099
N=60, cth=0.6, strong=1.0	both	697 / 1.18	488 / 1.57	307 / 1.05	\$7,117	52 / 1.83	\$22,322
N=60, cth=0.6, strong=None ◀ frozen	long	618 / 1.15	435 / 1.48	303 / 0.93	-\$9,047	47 / 1.13	\$3,947
N=60, cth=0.6, strong=None	short	591 / 0.87	379 / 1.01	240 / 1.03	\$3,490	44 / 1.60	\$13,849
N=60, cth=0.6, strong=None	both	1015 / 1.03	657 / 1.40	451 / 1.00	-\$44	70 / 1.35	\$15,205
N=60, cth=0.75, strong=1.0	long	1147 / 1.08	632 / 1.44	441 / 1.02	\$5,046	75 / 1.00	\$50
N=60, cth=0.75, strong=1.0	short	1175 / 1.08	624 / 1.05	406 / 1.16	\$35,626	88 / 1.05	\$3,443
N=60, cth=0.75, strong=1.0	both	1951 / 1.12	1020 / 1.36	706 / 1.14	\$52,451	131 / 1.16	\$16,386
N=60, cth=0.75, strong=None	long	1466 / 1.02	789 / 1.37	573 / 1.02	\$5,058	97 / 0.92	-\$6,263
N=60, cth=0.75, strong=None	short	1432 / 0.98	761 / 0.97	514 / 1.10	\$26,669	105 / 1.11	\$9,200
N=60, cth=0.75, strong=None	both	2352 / 1.02	1186 / 1.36	860 / 1.11	\$47,845	156 / 1.02	\$2,981
N=60, cth=None, strong=1.0	long	3474 / 0.94	1621 / 1.08	1204 / 1.06	\$45,604	241 / 1.03	\$7,146
N=60, cth=None, strong=1.0	short	3529 / 0.96	1605 / 1.03	1204 / 0.98	-\$16,606	251 / 0.98	-\$6,149
N=60, cth=None, strong=1.0	both	5455 / 0.95	2468 / 1.12	1867 / 1.07	\$89,407	374 / 0.94	-\$24,161
N=60, cth=None, strong=None	long	3849 / 0.94	1772 / 1.06	1347 / 1.05	\$39,627	263 / 0.99	-\$1,442
N=60, cth=None, strong=None	short	3820 / 0.95	1749 / 0.99	1309 / 0.98	-\$20,831	274 / 0.96	-\$12,761
N=60, cth=None, strong=None	both	5915 / 0.95	2636 / 1.12	2017 / 1.06	\$83,172	407 / 0.90	-\$45,623

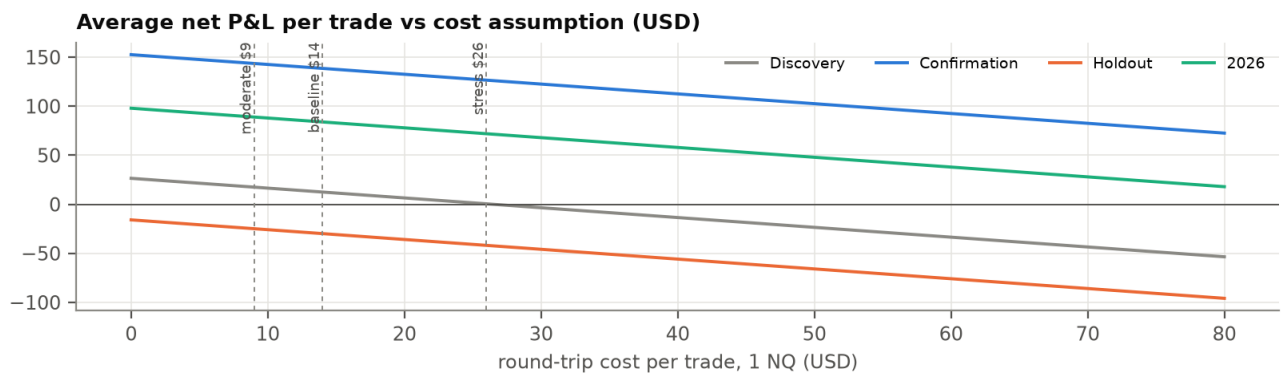
Neighbours with ≥ 20 trades and positive net: Discovery 16/36; Confirmation 31/36; Holdout 23/36; 2026 21/36

Stop / exit management. The frozen entry under every management variant from the campaign library (discovery stages A and B).

Stop / exit management	Disc. PF	Conf. PF	Holdout n / PF	Holdout net	2026 PF	2026 net
1×ATR stop; single target at +1R	0.82	1.15	469 / 0.86	-\$19,501	0.88	-\$3,378
1×ATR stop; single target at +1.5R	0.99	1.20	408 / 0.83	-\$24,637	1.01	\$179
1×ATR stop; single target at +2R	0.99	1.26	382 / 0.90	-\$12,948	0.96	-\$1,232
1×ATR stop; single target at +3R	1.07	1.36	358 / 0.86	-\$18,887	0.97	-\$833
1×ATR stop; single target at +4R	1.06	1.46	348 / 0.94	-\$8,322	1.03	\$725
1×ATR stop; no target, flat 16:00	1.10	1.57	330 / 0.87	-\$16,840	1.11	\$3,022
1×ATR stop; no target, exit after 60 min	0.93	1.22	429 / 0.77	-\$29,876	1.01	\$406
1.5×ATR stop; single target at +1R	1.04	1.17	389 / 0.96	-\$5,731	0.95	-\$1,525
1.5×ATR stop; single target at +2R	1.10	1.29	334 / 1.00	\$39	1.06	\$1,813

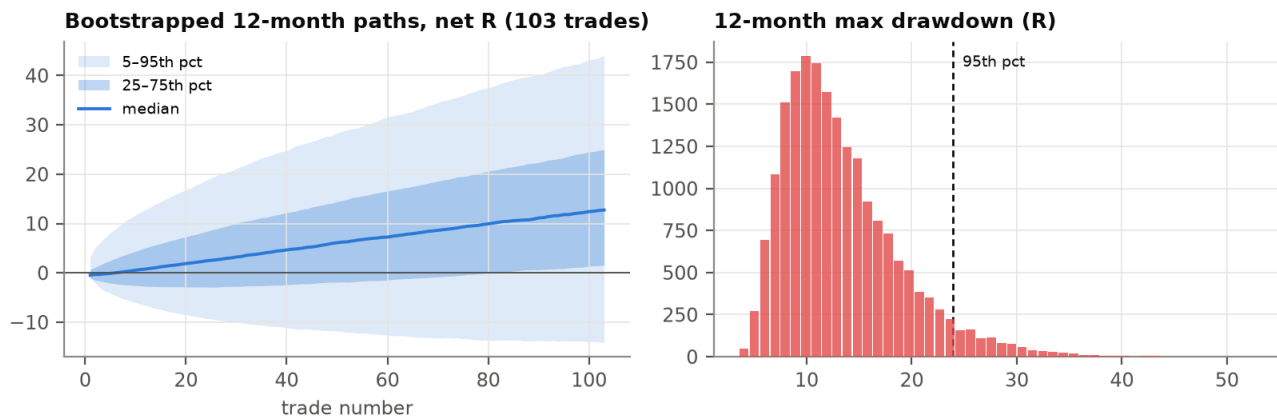
Stop / exit management	Disc. PF	Conf. PF	Holdout n / PF	Holdout net	2026 PF	2026 net
1.5×ATR stop; no target, flat 16:00 ◀ frozen	1.15	1.48	303 / 0.93	−\$9,047	1.13	\$3,947
5-min swing stop; single target at +1R	1.10	1.24	357 / 1.07	\$9,527	0.99	−\$285
5-min swing stop; single target at +2R	1.18	1.47	311 / 0.99	−\$1,059	1.15	\$4,549
5-min swing stop; no target, flat 16:00	1.16	1.46	286 / 0.96	−\$4,714	1.10	\$2,971
1×ATR stop; stop to entry at +0.5R, target +2R	0.66	1.02	429 / 0.68	−\$29,656	0.94	−\$1,275
1×ATR stop; stop to entry at +0.5R, no target	0.85	1.27	394 / 0.69	−\$26,761	0.90	−\$2,018
1×ATR stop; stop to entry at +1R, target +2R	0.91	1.18	388 / 0.80	−\$23,617	0.93	−\$1,757
1×ATR stop; stop to entry at +1R, no target	1.05	1.45	347 / 0.77	−\$24,758	0.94	−\$1,461
1×ATR stop; stop to entry at +1.5R, target +2R	0.94	1.27	383 / 0.83	−\$22,582	0.92	−\$2,082
1×ATR stop; stop to entry at +1.5R, no target	1.02	1.56	339 / 0.79	−\$25,176	0.90	−\$2,661
1×ATR stop; no target, 1.5×ATR trailing stop	1.06	1.33	356 / 0.82	−\$18,899	0.94	−\$1,419
1×ATR stop; no target, 2×ATR trailing stop	1.16	1.42	343 / 0.81	−\$21,812	0.87	−\$3,585
1×ATR stop; no target, 3×ATR trailing stop	1.11	1.48	334 / 0.84	−\$19,406	1.10	\$2,522
1×ATR stop; no target, trail beyond last 5-min swing	1.03	1.36	366 / 0.75	−\$29,254	0.90	−\$2,630
1×ATR stop; 50% at +1R, 50% at +2R	0.91	1.16	382 / 0.87	−\$14,798	0.88	−\$3,160
1×ATR stop; 50% at +1R, 50% at +3R	0.96	1.28	358 / 0.82	−\$20,360	0.90	−\$2,546
1×ATR stop; 50% at +1R, rest trails 2×ATR	1.00	1.28	342 / 0.80	−\$21,583	0.88	−\$3,000
1×ATR stop; 25% at +1R, 75% runs to 16:00	1.05	1.50	330 / 0.85	−\$17,372	1.04	\$1,004
1×ATR stop; 50% at +1R then stop to entry, rest +3R	0.90	1.21	370 / 0.78	−\$25,800	0.86	−\$3,638
1×ATR stop; stop to entry at +1R, 2×ATR trail	1.08	1.41	351 / 0.75	−\$27,499	0.77	−\$6,129

Cost sensitivity. The line shows what the average trade would be at any round-trip cost.



Break-even round-trip cost per contract (gross average trade): Discovery \$27; Confirmation \$153; Holdout n/a (gross loss); 2026 \$98. The baseline assumption is \$14.

H06-B (#18, long only): risk, sizing and forward expectations



Bootstrap of 785 trades from 2019–Aug 2026 (all out-of-sample relative to discovery), 103 trades per simulated year, 20,000 paths, trades drawn independently. This ignores regime persistence, so real drawdowns cluster more than shown.

Forward 12-month statistic (bootstrap)	In R	1 NQ at 2026 risk (1R ≈ \$1,538)	1 MNQ (1R ≈ \$154)
Median 12-month result	+12.7	\$19,584	\$1,958
5th percentile (bad year)	−14.1	−\$21,668	−\$2,167
95th percentile (good year)	+44.0	\$67,679	\$6,768
Probability of a losing year	22%		
Median max drawdown	12.1	\$18,617	\$1,862
95th pct max drawdown	23.9	\$36,780	\$3,678
99th pct max drawdown	30.8	\$47,379	\$4,738
Historical max drawdown 2010–26	44.8		

Suggested capital rule of thumb (not a margin requirement): hold at least the 99th-percentile 12-month drawdown plus exchange margin per unit traded. Dollar outcomes on one full NQ contract are far larger today than in most of the backtest; micro contracts (MNQ, 1/10 size) allow risk to be held near-constant per trade.

Portfolio context (25-strategy frozen cohort, 1 contract each)

Period	Corr. with rest of cohort (daily)	Median pairwise corr.	Most correlated members	Share of cohort P&L
holdout	0.10	0.01	#6 H06 both (0.43), #4 H47 long (0.16), #10 H08 long (0.15), #22 H47 long (0.09)	-1%
2026	0.32	0.13	#22 H47 long (0.55), #6 H06 both (0.42), #11 H07 long (0.34), #8 H08 long (0.33)	4%

Discussion

Was the selection a mistake? Not procedurally. Both variants met the preregistered confirmation criteria, and the cohort as a whole beat its holdout null (p 0.03). Out-of-sample decay of individual members is expected: 3 of 25 failed, and H06 accounts for 2 of the 3. What the record shows in hindsight is that H06's evidence came from one regime. In R terms discovery was close to zero, and about 37% of A's confirmation R came from 2022 alone.

Dollars vs R. A's holdout was $-\$4.4k$ but $+32R$. Its losses clustered on high-volatility days, where one contract risks more dollars, and its winners on quieter days. A constant-risk book would have made a modest profit (avg $+0.10R$, t 0.63), which is still not statistically distinguishable from zero. So the dollar failure is partly a sizing artefact, but the R result is not an edge either.

Where the residual edge sits (descriptive only). Across 2010–2026 the best trades fire in the first 30 minutes after 09:30 (A: $+0.79R$ per trade at 09:xx vs roughly zero later) and in the lowest-volatility third of days. After 2022, A's shorts averaged $+0.49R$ and its longs $-0.05R$. B, which is long only, was negative. These patterns were found after the results were known. They could motivate a *new* preregistered hypothesis tested on future data, but not a modification of this one.

Robustness. Across the 29 stop/exit variants, A's holdout was negative for 26 and B's for 27, so no management choice rescues it. In 2026 every one of A's 29 variants was positive. Taken together, that points to a regime-dependent effect rather than a stable edge.

Portfolio role. A is a genuine diversifier: holdout daily correlation with the rest of the cohort was only 0.22, and its median pairwise correlation 0.03. That is the only argument for keeping it under observation. It is not enough to trade a strategy with a failed holdout.

Recommendation

Recommendation: DO NOT TRADE either H06 variant. Both failed the preregistered, untouched 2023–2025 holdout, which is the test that counts. H06-B (long only) is negative on every post-freeze measure and should be dropped. H06-A stays in the frozen cohort's passive forward record only. It is not eligible for capital unless the true-forward record from 24 Sep 2026 independently meets the criteria on the last page.

H06-B (#18, long only): reject. It is negative in dollars and in R over 2023–2026 ($-\$5.1k$, $-18.5R$, 350 trades), failed the holdout well below its confirmation expectation (z -2.46), and no management variant fixes it.

H06-A (#6, long + short): do not allocate; observe only. It stays in the frozen cohort's true-forward record (it cannot be removed or changed under the protocol). It becomes eligible for capital only if the forward record, starting 24 Sep 2026, meets all of the conditions below on its own. The 2026 audit result must not be counted towards them.

Conditions, limits and kill criteria

- Forward evidence gate for H06-A: at least 150 true-forward trades (about 15 months at 120 per year), net positive after stress costs, and average $\geq +0.10R$ with $t \geq 1.5$.
- If the gate is ever met, start at micro size (MNQ) risking a fixed amount per trade, capped so a 41R drawdown (the bootstrap 99th percentile) is $\leq 5\%$ of the capital assigned to it.
- Kill switch (if ever live): drawdown beyond 45R, or a rolling-150-trade average below $-0.05R$.
- Any short-only or first-30-minutes variant must be preregistered as a new hypothesis and tested only on data after its registration date.

Caveats

- Backtests use 1-minute bars with pessimistic same-bar handling and fixed slippage. They do not model queue position, fast-market slippage beyond 2 ticks per side (stress case), or exchange outages.
- The 2026 sample (Jan–10 Aug) is short. The campaign's own record rates it low-power (cohort null p 0.26).
- The R-based view and the parameter/management grids in this report were computed after the holdout and 2026 results were known. They are diagnostic only and must not be used to modify or re-select the strategy. The preregistered process says the only valid new evidence now comes from the true-forward

record starting 24 Sep 2026.

- This is a quantitative research assessment of a systematic strategy, not personalised investment advice. Capital allocation decisions rest with the committee.

Appendix A: frozen specification (verbatim)

H06-A (#6, long + short)

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  "costs": {
    "baseline_round_trip_usd": 14.0,
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  },
  "direction": "both",
  "entry_conditions": {
    "down_event": "the close breaks below the previous 60-bar low; the previous 60 bars' realised vol was <= 0.6x its time-of-day norm; the 5-minute move >= 1.0 sigma units",
    "mode": "continuation",
    "rule": "continuation: BUY on up-event, SELL on down-event; direction=both",
    "up_event": "the close breaks above the previous 60-bar high; the previous 60 bars' realised vol was <= 0.6x its time-of-day norm; the 5-minute move >= 1.0 sigma units"
  },
  "entry_id": "Efc242ab81ddb80b3",
  "family": "v3_h06_compression_expansion",
  "instrument": "NQ",
  "management": {
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    "be_R": null,
    "be_after_leg": null,
    "exit_opp": false,
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    "needs_stop": true,
    "params": {
      "hold": "eod"
    },
    "trail": null
  },
  "parameters": {
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    "cth": 0.6,
    "mode": "continuation",
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  "trading_window": {
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    "entry_start": "09:30",
    "flat_at": "16:00",
    "name": "rth"
  }
}
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H06-B (#18, long only)

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{
  "costs": {
    "baseline_round_trip_usd": 14.0,
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  },
  "direction": "long",
  "entry_conditions": {
    "down_event": "the close breaks below the previous 60-bar low; the previous 60 bars' realised vol was <= 0.6x its time-of-day norm",
    "mode": "continuation",
    "rule": "continuation: BUY on up-event, SELL on down-event; direction=long",
    "up_event": "the close breaks above the previous 60-bar high; the previous 60 bars' realised vol was <= 0.6x its time-of-day norm"
  },
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  "family": "v3_h06_compression_expansion",
  "instrument": "NQ",
  "management": {
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    "be_R": null,
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    "exit_opp": false,
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    "kind": "stop_time",
  }
```

```

"legs": [
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],
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"params": {
  "hold": "eod"
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"trail": null
},
"parameters": {
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  "cth": 0.6,
  "mode": "continuation",
  "strong": null
},
"stop": {
  "method": "atr",
  "mult": 1.5
},
"strategy_id": "C39c52e02a3c54412",
"trading_window": {
  "entry_end": "15:30",
  "entry_start": "09:30",
  "flat_at": "16:00",
  "name": "rth"
}
}

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Appendix B: provenance

Item	Value
Final cohort freeze	freezes/FINAL_COHORT_FREEZE.json · sha256 f5666dac728aa2539b2f8223309012914d71923ab71f3363e9e035344ace8f03
Frozen (UTC)	2026-09-22T23:21:51.429326+00:00
Discovery freeze	sha256 0343a974953d53dc026fd1f7ef5c124c8985213b132fba5df4f8a986fc85b2c9
Master record	results/V3_MASTER_RESEARCH_RECORD.md
Sealed holdout / audit trades	results/holdout/trades.parquet · results/audit2026/trades.parquet
This report	reports/H06_Compression_to_Expansion_Research_Report.pdf, built from trades regenerated with the frozen specs (exact match with the sealed records) + stored discovery/confirmation evidence
Known erratum	results/audit2026/ERRATUM_2026-09-22.json (X3 null p only; does not affect this report)